

LAB ON OBJECT ORIENTED PROGRAMMING

EXPERIMENTS:

Implement the programs with WINDOWS / LINUX platform using OOPS

Paper : 21CAC-4P/SD Theory/Week: 4 Hours Credits: 2	LAB ON OOP	Hrs: 40 IA : 25 Exam: 25
Course Objectives: 1. To enable the students to learn the skills in object-oriented programming concepts 2. To impart the working of C++ classes and class libraries 3. To implement the concept of arrays, vectors, inheritance and file I/O stream concepts in C++		
Course Outcome: After the completion of the course, the students will be able to CO1: To relate the concepts of programming in C and C++. CO2: To demonstrate Object Oriented Programming Concepts in C++. CO3: To create a application using all the concepts of C++		
1.	Write a C++ program to create a class with data members like principal amount, time and rate of interest. Create a member function to accept data values, to compute simple interest and to display the result. https://www.youtube.com/watch?v=XtCQrObyHSI	
2.	Write a C++ program to perform the addition and subtraction of two complex numbers using member functions. https://www.youtube.com/watch?v=GWZrcyfl5qc	
3.	Write a C++ program to calculate the volume of cube, cylinder and cuboid using function overloading. https://www.youtube.com/watch?v=i9W49kRQ3ec	
4.	Write a C++ Program to create a class 'distance' with the data members feet and inches and member functions read() and show(). Write a c++ program to add two distances by using object asfunction argument.(inches should not be >=12) https://www.youtube.com/watch?v=bdBbWdofzVo	
5.	Write an object oriented program in C++ to create a library information system containing the following for a book in the library: Accession number, Author name, Title of the book, Year of publication, Publisher's name, Cost of the book. Define a class with data members and suitable member functions for initializing and for destroying the data via constructor and destructor. https://www.youtube.com/watch?v=oBZzr-0IO_k	

6.	Create a class rectangle with length, breadth and area. Create another class cuboid that inherits rectangle and has additional members height and volume. Use single inheritance property. https://www.youtube.com/watch?v=G4olQafKERg
7.	Create a class containing the following data members Register No, Name and
	Fees. Also create a member function to read and display the data using the concept of pointers to objects. https://www.youtube.com/watch?v=UwvUVb_EUUo
8.	Write a C++ program to add, subtract, multiply and divide two numbers using class template. https://www.youtube.com/watch?v=jXWRzkHHcg
9.	Write a C++ program to create a class with data members a, b, c and member functions to input data, compute discriminants based on the following conditions and print the roots. If discriminant = 0, print the roots are equal and their value. If discriminant > 0, print the real roots and their values. If discriminant < 0, print the roots are imaginary and exit the program. https://www.youtube.com/watch?v=Hvgay40lndo
10.	Write a C++ program to compute the total marks and declare the results using array of objects. Assume that the class contains the data members - roll no, name, marks in 3 subjects. Result is calculated as follows. If student gets <35 fail. Otherwise various results are calculated on average as >=70 Distinction. >=60 and <70 First Class. >=50 and <60 Second Class else Pass Class https://youtu.be/HNkfKTMFD4E
11.	Create a class 'Bank' which includes data members—Acno, Name, Balance and parameterized constructor to initialize the data members and other methods like deposit, withdrawal, and display the detail of the customer. (Note: Minimum balance of Rs. 500/- should be maintained.) https://www.youtube.com/watch?v=rr04LBiz19U
12.	Using constructors and proper methods design a class graphics which stores shapes, area, back color and fore colors. Use this class in the main program to input any 'N' shapes and perform the following options and print the list in the neat format. 1. Sort according to Area. b. Search according to accepted shape. https://youtu.be/dLl8BfrNjrM
13.	Write a program to accept two strings and using operator overloading perform the following. a. Concatenation of two strings. b. Comparison of two strings alphabetically.

14.	Create a class called 'Time' which includes the data members – hours, minutes and seconds. Use the method ▪ To accept the time. ▪ To display the time. ▪ To increment time by one second by overloading unary operator ++ ▪ To decrement time by one second by overloading unary operator - - Write a menu driven program for the above operation, (Hint: Minutes and Seconds must be always within the range 0-59).
15.	Multiple Inheritance: Write a program to create a class 'Personnel Information' which includes name, address and gender as the data members. Another class for 'Physical Information' with data members height, weight, blood group. Derive a class called 'Salary' with employee number, department and salary. Find increment in salary for an employee as follows. For Male: In department S or P- 10% For Female: In department S or P- 11%. (Hint: S for Sales and P for Purchase, for any other department no increment) https://youtu.be/S40DPqAt8zo

Continuous Internal Assessment (CIA) Method:

Sl. No	Type of Assessment	Mode of Assessment	Marks
1	Observation Book	Regular mode of Assessment	5
2	Lab Internal	Regular mode of Assessment	10
3	Record	Regular mode of Assessment	5
4	Attendance		5
Total			25

SCHEME OF EXAMINATION FOR END SEMESTER EXAMINATION OF 50 MARKS:

Two experiments, one from each part are to be performed by the students in the examination.

Sl. No	Type of Assessment	Mode of Assessment	Marks
1	Part –A	Regular mode of Assessment	10
2	Part –B	Regular mode of Assessment	10
3	Viva	Regular mode of Assessment	5
Total			25